Domestic and Commercial Electricity Bill Engine with Slab Tariff and PF Penalty

Design Document | ECE-023 | Capstone Project

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Section	ECM B
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Domain	Power & Energy - Distribution Billing
Language	Python 3.10+ (standard library only)
Repository	E035-ECE-023
Document version	1.1 (updated to the official Week 2 sample data pack)

Every business rule is registered as BR-nn, every validation as VR-nn. Section 11 maps each BR to its code and tests.

1. Purpose and scope

The engine turns monthly meter readings into bills for domestic and small commercial consumers of a BESCOM-style distribution company. It is driven by a command file (no UI). In scope: consumer registration, reading ingestion, monthly bill generation with line items, payment recording, late-payment interest carried to the next bill, and a monthly collection summary per category. Out of scope: GUI, database, real-time metering, HT/industrial tariffs, subsidies (stretch goal; the design leaves a hook - see section 12).

Success criterion (from the brief): for the given consumer, reading and payment files, bills.csv and collection_summary.csv match the expected output field-for-field. Therefore: all money is `Decimal`, rounding is explicit and half-up, output formatting is deterministic, and diagnostics never pollute the result stream.

2. Business rule register

BR-01 to BR-08 are the rules stated in the brief, in the brief's order. BR-09 to BR-14 are **derived rules**: the brief is silent or ambiguous, so the decision is explicit (reasoning in DECISION_LOG.md). The official sample data pack (Week 2) stated its conventions for these points; where the pack settles a rule, the engine follows the pack.

ID	Rule	Precise statement used by the engine
BR-01	Domestic slabs (telescopic)	Units 1-50 @ Rs 4.75; 51-100 @ 7.00; 101-200 @ 8.25; above 200 @ 9.50. Each unit is priced by the slab it falls in.
BR-02	Commercial slabs	Units 1-100 @ Rs 9.00; above 100 @ 10.50 (also telescopic).
BR-03	Fixed charge	Sanctioned kW x Rs 110 (DOMESTIC) or Rs 160 (COMMERCIAL), every month, even at 0 units.
BR-04	PF penalty	COMMERCIAL only. PF < 0.90: 1% of energy charge per full 0.01 below 0.90, cap 15%. PF >= 0.90: none. (PF 0.70 = 20 steps -> capped at 15%.)
BR-05	Electricity tax	9% of (energy + fixed + PF penalty), rounded to 2 dp. Late interest and rounding are not taxed.
BR-06	Meter rollover	5-digit meter. closing >= opening: units = closing - opening. closing < opening: units = (99999 - opening) + closing + 1.
BR-07	Late payment	Due date = bill date + 15 days; a payment ON the due date is on time. Payment after the due date: interest = 1% x bill total x ceil(days late / 30), 2 dp, shown as the late-interest line of the consumer's FIRST bill dated strictly AFTER the payment date. Interest is not taxed.
BR-08	Rounding	Each line item 2 dp (half-up). Bill total rounded to nearest rupee (half-up); the difference is its own line item and may be negative.
BR-09	Bill date and month	Bill date = reading date; bill month = YYYY-MM of the reading date; at most one bill per consumer per month (pack convention).
BR-10	Opening reading chain	Opening = meter_start_reading for the first bill, then the previous closing reading. Equal readings = zero units, not a rollover (pack convention).
BR-11	Late measure	Each payment pays one bill in full. Days late = payment date - due date. 30 days = 1 period, 31 days = 2 periods.
BR-12	Interest charged once	A late payment's interest is placed on exactly one bill and never repeated. Base = total of the paid bill (including any interest that bill carried). Unpaid bills attract no interest.
BR-13	Payment validation	A payment must name an existing bill, must not pre-date the bill, must not exceed the outstanding amount, and needs a unique payment id; an amount different from the bill total is accepted with a WARN.
BR-14	Collection summary	Per bill month and category (DOMESTIC first): bills, units, billed, collected (any payment date), outstanding = billed - collected, late_payments = bills paid after due date, interest_billed = late-interest lines on that month's bills.
BR-15	Output files	bills.csv (one row per consumer per month, ordered by month then RR) and collection_summary.csv, in the exact column order of the pack; readings keep 5 digits; money as plain 2-dp decimals.

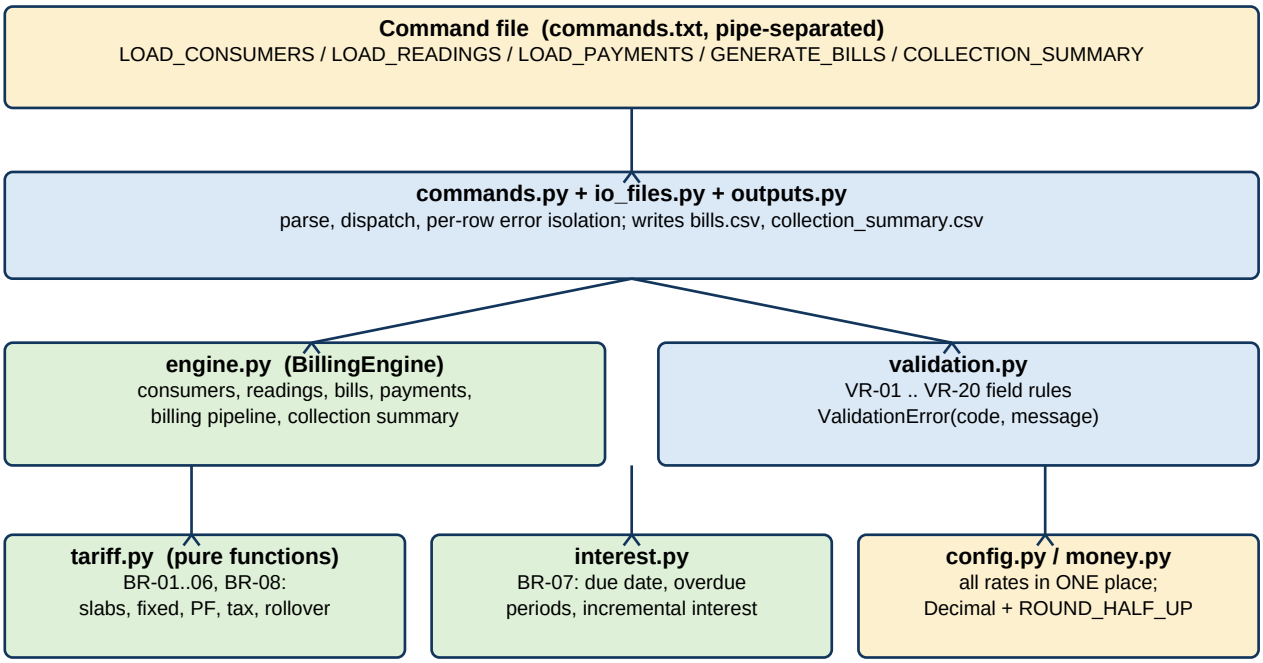
3. Domain model

Entity	Key attributes	Notes
Consumer	rr (key), category, load_kw, start_reading; state: last_closing, last_reading_date, last_billed_month	State advances only when a bill is generated successfully.
Reading	rr, reading_date, closing, pf	Unique per (rr, month). PF mandatory for COMMERCIAL, ignored for DOMESTIC.

Entity	Key attributes	Notes
Bill	bill_id = rr-YYYY-MM, bill/due date, opening, closing, units, rollover flag, slab lines, energy, fixed, pf_penalty, tax, interest (+ per-bill detail), rounding, total	Immutable once generated, except its payment ledger and interest_levied counter.
Payment	payment_id, rr, bill_month, pay_date, amount	Attached to exactly one bill; held as pending if that bill does not exist yet.
SlabLine	from_unit, to_unit, units, rate, amount	Printed on the bill so a customer (and the examiner) can audit the energy charge.

Relationships: Consumer 1-N Reading (one per month); Consumer 1-N Bill (one per month); Bill 1-N Payment; Bill 1-N interest contributions to later bills of the same consumer.

4. Architecture



Design principles. (1) Rules live in pure functions (tariff.py, interest.py) with no state or I/O, so each rule is unit-tested and can be explained alone. (2) All rates are in config.py - tariff changes touch one file. (3) Commands are thin: they parse and delegate to the engine. (4) A bad row or command never aborts a run: it is reported with file/line and skipped. (5) Results go to stdout, diagnostics to stderr, exit code 1 if any ERROR - so stdout can be diffed against the expected output.

5. Processing pipeline for one bill

Step	Action	Rule
1	Locate the consumer's reading for the requested month; skip (WARN) if absent or already billed	BR-09
2	Check month is after the last billed month	VR-15
3	units = f(opening, closing) with rollover detection	BR-06, BR-10
4	Energy charge by telescopic slabs (slab lines retained)	BR-01/02
5	Fixed charge = kW x rate	BR-03
6	PF penalty (commercial) from energy charge and PF	BR-04
7	Tax = 9% x (energy + fixed + PF penalty)	BR-05
8	Late interest: every not-yet-charged late payment of this consumer's earlier bills dated before this bill date adds 1% x paid-bill total x periods	BR-07/11/12
9	gross = sum of lines; total = round-half-up to rupee; rounding = total - gross	BR-08
10	Commit: update consumer state, store bill, mark interest as levied, attach any pending payments	-

6. Key algorithms

6.1 Telescopic slabs

`lower = 0; for each (upper, rate): if units <= lower stop; top = min(units, upper); n = top - lower; amount = n x rate; lower = top`. Because *lower* advances slab by slab, each unit is priced exactly once at the rate of the slab it occupies. Boundary behaviour: 50 units stays entirely in slab 1 (Rs 237.50); 51 units adds exactly one unit at Rs 7.00 (Rs 244.50).

6.2 Meter rollover

If closing < opening the meter has passed 99999 and restarted at 00000. The units are those from opening up to 99999 (*99999 - opening*), plus the step from 99999 to 00000 (*+1*), plus the new reading (*closing*). Example from the brief: opening 99980, closing 00012: 19 + 1 + 12 = **32 units**. Closing equal to opening is zero consumption, not a full wrap.

6.3 PF penalty

steps = floor((0.90 - PF) / 0.01); penalty% = min(steps x 1%, 15%); penalty = energy x penalty%. All arithmetic in Decimal, so PF 0.89 is exactly one step (no floating-point 0.8999.. surprise). PF 0.90 gives zero steps; PF 0.70 gives 20 steps and is capped to 15%.

6.4 Late-payment interest

Interest is driven by **late payments**, not by unpaid balances. For a payment made after its bill's due date: periods = ceil(days late / 30); interest = 1% x *total of the paid bill* x periods, rounded to 2 dp. The amount is placed on the consumer's **first bill whose bill date is strictly after the payment date** and is never charged again. It is not taxed. A bill that is still unpaid (for example an October bill when the data stops on 31 Oct) attracts no interest.

Payment of a Rs 1270 bill (due 12 Aug)	Days late	Periods	Interest	Charged on
12 Aug (on the due date)	0	0	Rs 0.00	-
13 Aug (one day after)	1	1	Rs 12.70	next bill dated after 13 Aug
11 Sep	30	1	Rs 12.70	next bill dated after 11 Sep
12 Sep	31	2	Rs 25.40	next bill dated after 12 Sep
Never paid	-	-	Rs 0.00	-

7. Input formats and validation rules

File	Header (case-insensitive)
consumers.csv	rr_number, consumer_name, category, sanctioned_load_kw, meter_start_reading, area, mobile (extra columns ignored)
readings.csv	rr_number, reading_date (YYYY-MM-DD), closing_reading (5 digits, leading zeros kept), avg_pf (NONE for DOMESTIC; 0.00 is legal for COMMERCIAL)
payments.csv	payment_id, rr_number, bill_month (YYYY-MM), payment_date, amount_rs, mode

Code	Rule	Code	Rule
VR-01	RR number like RR-BG-50108 (letters, digits, hyphens)	VR-11	Month is YYYY-MM
VR-02	Category is DOMESTIC or COMMERCIAL	VR-12	Payment amount > 0, max 2 dp
VR-03	Sanctioned load > 0, max 2 dp, finite	VR-13	RR number is registered
VR-04	Meter start reading: whole number, 0-99999	VR-14	One reading per consumer per month
VR-05	Reading date is a real calendar date	VR-15	Bill month after last billed month
VR-06	RR number not already registered	VR-16	Reading date after previous reading date
VR-07	Closing reading whole number, 0-99999	VR-17	Payment id unique
VR-08	PF mandatory for COMMERCIAL (NONE/blank rejected)	VR-18	Payment not before bill date
VR-09	PF in [0, 1]	VR-19	Payment <= outstanding amount
VR-10	Payment / as-of date is a real date	VR-20	Required CSV columns present

Code	Rule	Code	Rule
CMD-01	Command is known	CMD-02	Correct argument count

8. Edge cases mapped to evidence

Edge case (brief + pack)	Handling	Automated test	Pack location
50 / 100 / 200 units; 201	Telescopic loop; boundary stays in lower slab	test_exactly_50/100/200_units, test_201_units	readings lines 6, 59, 95; 53
PF exactly 0.90	0 steps -> no penalty	test_pf_exactly_090_no_penalty	line 97 RR-BK-21066
PF 0.70 (cap)	20 steps -> capped at 15%	test_pf_070_cap_15_percent (1647.00)	line 13 RR-JP-78161
PF 0.89	1 step = 1%	test_pf_089_one_percent	line 2 RR-BG-50108
Rollover 99980 -> 00012 (+ second)	(99999-99980)+12+1 = 32; 99731 -> 00285 = 554	test_rollover_99980_to_00012, test_second_rollover	lines 75, 92
Zero-consumption months (4)	Energy 0; fixed + tax still billed; PF 0.00 accepted	test_zero_consumption_months	lines 10, 29, 52, 118
Paid on due date vs one day late	0 days vs 1 day late	test_pay_0035..., test_pay_0036...	payments lines 36, 37
Other late payments (30 / 31 / 44 days)	ceil(days/30) periods; lands on first later bill	test_pay_0070, 0078, 0080, 0079	payments lines 71, 79, 81, 80
Rounding at exactly 50 paise	Half-up -> +0.50	test_rounding_exactly_50_paise_half_up	line 66 RR-JP-94514
Fractional load; unpaid October bills	kW x rate; outstanding, no interest	test_fractional_load, test_unpaid_october...	-

9. Worked bills (hand-verifiable)

These are the bills the engine prints for the inputs below; every figure can be reproduced with a calculator.

9.1 Domestic, 3 kW, opening 00120 -> closing 00250 (130 units)

Step	Working	Result
Units	250 - 120	130
Slab 1	50 x 4.75	237.50
Slab 2	50 x 7.00	350.00
Slab 3	30 x 8.25	247.50
Energy charge	237.50 + 350.00 + 247.50	835.00
Fixed charge	3 kW x 110	330.00
Tax	9% x (835.00 + 330.00 + 0.00) = 104.85	104.85
Gross	835.00 + 330.00 + 0.00 + 104.85	1269.85
Rounding line	1270 - 1269.85	0.15
Total payable	nearest rupee	**Rs 1270**

Telescopic vs flat. A *flat* slab prices all 130 units at the rate of the slab the total lands in (Rs 8.25): 130 x 8.25 = Rs 1072.50. *Telescopic* prices only the units above each threshold at the higher rate: Rs 835.00. The telescopic method has no cliff - going from 100 to 101 units adds only Rs 8.25 (Rs 587.50 -> Rs 595.75), whereas a flat scheme would suddenly re-price all 101 units at 8.25 (Rs 833.25). That is why consumption exactly at a boundary is a named edge case.

9.2 Commercial, 5 kW, 150 units, average PF 0.85

Step	Working	Result
Energy	100 x 9.00 + 50 x 10.50 = 900.00 + 525.00	1425.00
Fixed	5 kW x 160	800.00
PF penalty	0.90 - 0.85 = 5 steps -> 5% x 1425.00	71.25
Tax	9% x (1425.00 + 800.00 + 71.25) = 206.6625	206.66
Gross / rounding	2502.91 -> 2503	0.09
Total payable		**Rs 2503**

9.3 Rollover and zero-consumption

Case	Working	Energy / Fixed / Tax	Total
Domestic 1 kW, 99980 -> 00012	units = 19 + 12 + 1 = 32; 32 x 4.75	152.00 / 110.00 / 23.58	Rs 286
Domestic 2 kW, 500 -> 500	0 units; fixed 2 x 110; tax 9% x 220.00	0.00 / 220.00 / 19.80	Rs 240

9.4 Late interest carried to the next bill

The Rs 1270 bill of 28 Jul 2026 (due 12 Aug) is paid on 13 Aug: 1 day late = 1 period, so interest = 1% x 1270 = **Rs 12.70**. The first bill dated after 13 Aug is the 28 Aug bill, which therefore shows: energy 0.00 + fixed 330.00 + PF 0.00 + tax 29.70 + interest 12.70, rounded -> **Rs 372** (interest is not taxed).

Printed bill (July, domestic 130 units):

```
================================================================
                   ELECTRICITY BILL - 2026-07
================================================================
Bill No         : RR1001-2026-07
RR No           : RR1001              Category : DOMESTIC
Sanctioned load : 3.00 kW
Bill date       : 2026-07-28   Due date : 2026-08-12
Meter reading   : opening 00120  closing 00250  units 130
----------------------------------------------------------------
  Units     1 - 50       50 x Rs 4.75                    237.50
  Units    51 - 100      50 x Rs 7.00                    350.00
  Units   101 - 130      30 x Rs 8.25                    247.50
Energy charge                                            835.00
Fixed charge (3.00 kW x Rs 110)                          330.00
PF penalty (not applicable)                                0.00
Electricity tax (9%)                                     104.85
Late payment interest                                      0.00
Rounding adjustment                                        0.15
----------------------------------------------------------------
TOTAL PAYABLE (Rs)                                      1270.00
================================================================
```

10. Test strategy

Layer	What it proves	Where
Unit - rules	Each BR in isolation incl. every boundary (0,1,50,51,100,101,200,201), PF 0.89/0.895/0.90/0.75/0.70, rollover variants, half-up rounding	tests/test_tariff.py
Unit - validation	Every VR code is reachable and rejects bad input; NONE / 0.00 PF handling	tests/test_validation.py
Engine	Bill assembly, reading chain, late-interest rules (on due date, +1, +30, +31 days, lands after payment date, charged once, untaxed), payment validation, summary	tests/test_engine.py
Official pack	Every planted scenario by README line number; 120 bills vs independent oracle; all 108 pack payments equal engine totals; CSV formats; golden snapshots; second independent implementation (standalone/) gives byte-identical files; bad-input robustness	tests/test_end_to_end.py

Run: `python -m unittest discover -s tests -t .` (no third-party packages). Invariants asserted on every bill: lines add up to the total; total is a whole rupee; |rounding| <= 0.50; summary billed = sum of bills; collected + outstanding = billed. Strongest evidence: the pack's payment amounts were generated from the official totals, and all 108 equal our computed totals.

11. Traceability: business rule to code to tests

Rule	Implementation	Tests
BR-01/02	tariff.energy_charge, config.SLABS	TestDomesticSlabs, TestCommercialSlabs
BR-03	tariff.fixed_charge	TestFixedCharge, test_zero_consumption_month
BR-04	tariff.pf_penalty	TestPFPenalty, test_commercial_pf_070_cap
BR-05	tariff.electricity_tax	TestTax
BR-06	tariff.units_consumed	TestRollover, test_rollover_month_from_brief
BR-07/11/12	interest.periods_overdue / interest_for_payment; engine._bill_one	TestLateInterest (10 tests), test_pay_0035..0080
BR-08	money.q2 / round_rupee; engine._bill_one	TestRounding, test_line_items_always_reconcile_to_total
BR-09/10	engine._bill_one, Consumer state	test_reading_chain_uses_previous_closing, test_rebilling_month_is_idempotent
BR-13	engine.record_payment / _attach	TestPaymentValidation
BR-14/15	engine.collection_summary, outputs.py	TestSummary, TestPackOutputFiles

12. Assumptions, risks and extension points

Resolved by the official pack. Bill date = reading date; PF steps are whole 0.01 steps; late interest = 1% x bill total x ceil(days late/30) on the first bill dated after the payment; summary definitions; output file formats. **Still our own decisions** (pack silent): payments that pre-date their bill or exceed its total are rejected, a differing amount only warns; a consumer with no reading in a month gets no bill and no fixed charge; bad rows are skipped and reported.

Risk: hidden month. The third month's expected output is hidden, so correctness must come from rule fidelity, not from fitting visible data. Mitigations: independent oracle test, a second independent implementation, boundary tests, every ambiguity written down before coding.

Extension points (stretch goals). Subsidy for consumers under 40 units: add a rule function between energy charge and tax and a line item in Bill/report. Arrears ageing 30/60/90: read-only report over Bill.outstanding and due_date - no change to billing logic.

Limitations. A closing reading lower than opening is always treated as rollover (a meter replacement or data-entry error is indistinguishable); such bills carry an explicit rollover note so a human can review them.